

IV.POS.8 D1.B — CGC Coarsening Variant

Corpus: 20 decay-comparison DBs (10 R2 V5 + 5 D1.A decayexp + 5 D1.A decayep

Variants: `production_log2_corrected`, `region_only`, `log4_explicit`, `page_`

All memory keys from **corrected replay** (Batch 1.6); lookup pass-through from `hook3_raw`.

Artifacts from `build_d1b_artifacts.py`. Plots under `plots/`.

```
In [1]: import json, sqlite3, sys
        from pathlib import Path
        import matplotlib
        matplotlib.use('Agg')
        import matplotlib.pyplot as plt
        import pandas as pd
        import numpy as np
        from IPython.display import Image, display, Markdown

        REPO = Path('/root/arguzz')
        D1B = REPO / 'a4/runs/iv_pos_8/d1b'
        IV7 = REPO / 'a4/runs/iv_pos_7'
        sys.path.insert(0, str(REPO))
        sys.path.insert(0, str(IV7))
        sys.path.insert(0, str(D1B / 'analysis'))

        from analysis.metrics import N_MUTATIONS, _coverage_curve
        from d1b_cgc_maps import VARIANTS, variant_first_hit_map
        from replay_cgc_corrected import replay_memory_first_hits, replay_lookup_

        PLOTS = D1B / 'plots'
        PLOTS.mkdir(exist_ok=True)
        COLORS = {'V5': '#1f77b4', 'V5-decayexp': '#2ca02c', 'V5-decayepoch': '#d

        metrics = pd.read_csv(D1B / 'd1b_metrics_table.csv')
        paired = pd.read_csv(D1B / 'd1b_paired_tests.csv')
        saturation = pd.read_csv(D1B / 'd1b_saturation_profile.csv')
        summary = json.loads((D1B / 'd1b_build_summary.json').read_text())
        manifest = (D1B / 'd1b_artifacts.sha256').read_text()

        def show_plot(fig, path):
            fig.savefig(path, dpi=120, bbox_inches='tight')
            plt.close(fig)
            display(Image(filename=str(path)))

        def cgc_curve(db_path, cgc_variant):
            with sqlite3.connect(db_path) as conn:
                mem = replay_memory_first_hits(conn)
                lookup = replay_lookup_first_hits(conn)
                fhm = variant_first_hit_map(cgc_variant, mem, lookup, conn)
                return _coverage_curve(sorted(fhm.values()), n=N_MUTATIONS)
```

Headline numbers (V5 mean cgc_final)

```
In [2]: rows = []
for cv in VARIANTS:
    for corpus in ('V5', 'V5-decayexp', 'V5-decayepoch'):
        m = metrics[(metrics['cgc_variant']==cv) & (metrics['corpus_variant']==corpus)]
        rows.append({'cgc_variant': cv, 'corpus_variant': corpus, 'mean_cgc': m['mean_cgc'].mean())
display(pd.DataFrame(rows).pivot(index='cgc_variant', columns='corpus_variant'))
```

	corpus_variant	V5	V5-decayepoch	V5-decayexp
	cgc_variant			
	log4_explicit	120.6	115.8	114.2
	page_class	110.7	105.8	104.2
	production_log2_corrected	218.2	218.0	215.6
	region_only	105.7	100.8	99.2

Saturation overlay vs local catalog saturation (paired V5)

```
In [3]: # Overlay: mean cumulative CGC curves (4 variants, V5 paired seeds) vs local catalog saturation
from analysis.discover import discover_dbs

VARIANT_COLORS = {
    'production_log2_corrected': '#1f77b4',
    'region_only': '#d62728',
    'log4_explicit': '#ff7f0e',
    'page_class': '#2ca02c',
}

VARIANT_LABELS = {
    'production_log2_corrected': 'production_log2_corrected',
    'region_only': 'region_only',
    'log4_explicit': 'log4_explicit',
    'page_class': 'page_class',
}

paired_seeds = summary['paired_seeds']
x = np.arange(1, N_MUTATIONS + 1)
m_v5 = discover_dbs(IV7 / 'dbs', variants=('V5',))

# Local saturation anchor: mean time_to_46 on paired V5 seeds (from D1.A)
import pandas as pd
d1a = pd.read_csv(REPO / 'a4/runs/iv_pos_8/d1a/d1a_metrics_table.csv')
local_sat_mean = float(d1a[(d1a.variant=='V5') & (d1a.seed.isin(paired_seeds['seed'])]['time_to_46'].mean())
print(f'Local saturation anchor (mean time_to_46): {local_sat_mean:.0f}')

curves_by_variant = {}
for cv in VARIANTS:
    curves = []
    for seed in paired_seeds:
        db = m_v5['V5'][seed]
        curves.append(cgc_curve(db, cv))
    curves_by_variant[cv] = np.array(curves)
```

```

# --- Absolute overlay ---
fig, ax = plt.subplots(figsize=(11, 6))
for cv in VARIANTS:
    arr = curves_by_variant[cv]
    mean_curve = arr.mean(0)
    ax.plot(x, mean_curve, color=VARIANT_COLORS[cv], lw=2.5, label=VARIANT_COLORS[cv])
    row = saturation[saturation.cgc_variant == cv].iloc[0]
    sat_mut = int(row.saturation_mutation_id)
    sat_idx = min(sat_mut, N_MUTATIONS) - 1
    ax.scatter([sat_mut], [mean_curve[sat_idx]], color=VARIANT_COLORS[cv],
               edgecolors='black', linewidths=1.2, marker='o')

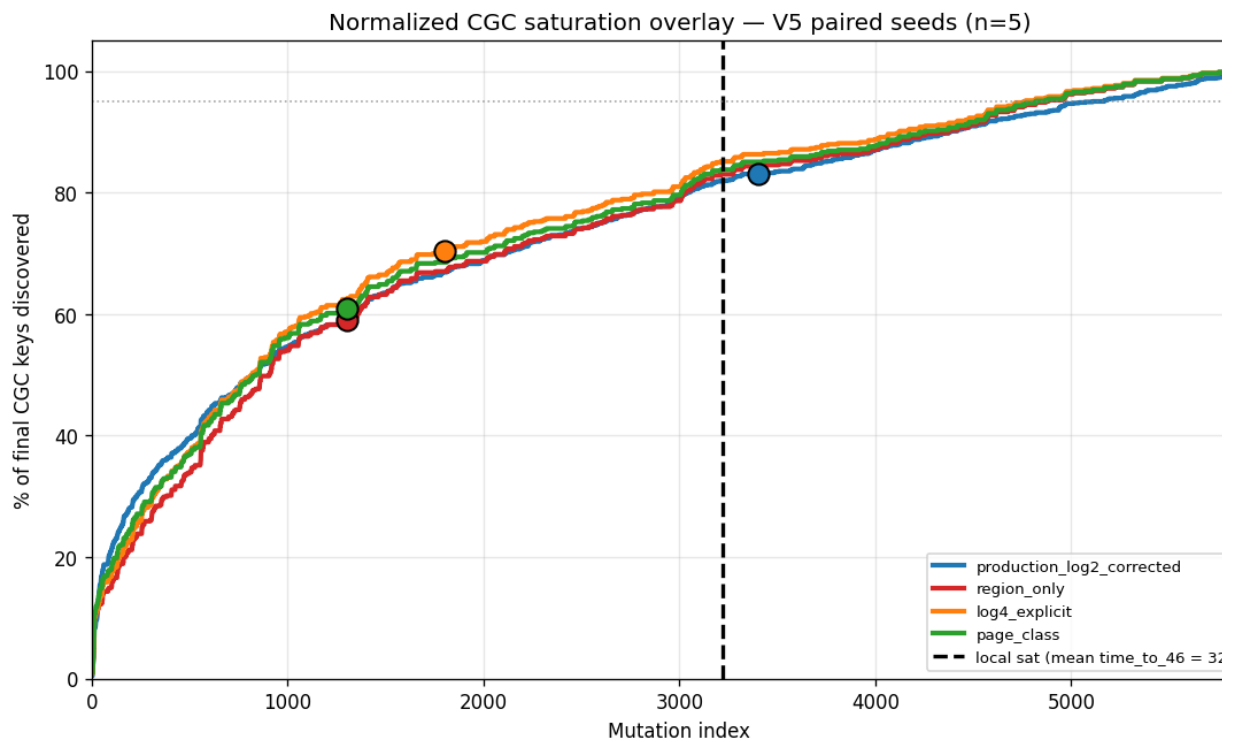
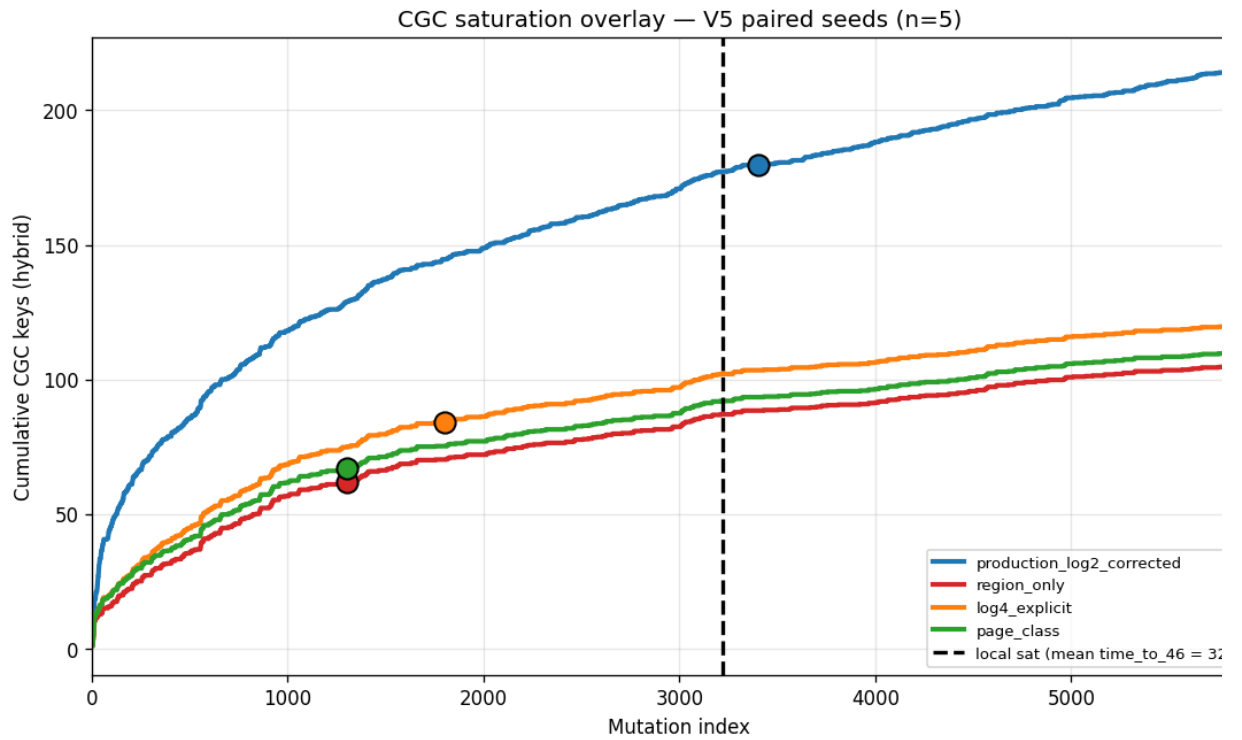
ax.axvline(local_sat_mean, color='black', ls='--', lw=2,
           label=f'local sat (mean time_to_46 = {local_sat_mean:.0f})')
ax.set_xlabel('Mutation index')
ax.set_ylabel('Cumulative CGC keys (hybrid)')
ax.set_title('CGC saturation overlay – V5 paired seeds (n=5)')
ax.legend(fontsize=8, loc='lower right')
ax.grid(alpha=0.3)
ax.set_xlim(0, N_MUTATIONS)
show_plot(fig, PLOTS / 'd1b_saturation_overlay_v5.png')

# --- Normalized overlay ---
fig, ax = plt.subplots(figsize=(11, 6))
for cv in VARIANTS:
    arr = curves_by_variant[cv]
    finals = arr[:, -1].astype(float)
    finals[finals == 0] = np.nan
    norm = arr / finals[:, None] * 100.0
    mean_norm = np.nanmean(norm, axis=0)
    ax.plot(x, mean_norm, color=VARIANT_COLORS[cv], lw=2.5, label=VARIANT_COLORS[cv])
    row = saturation[saturation.cgc_variant == cv].iloc[0]
    sat_mut = int(row.saturation_mutation_id)
    sat_idx = min(sat_mut, N_MUTATIONS) - 1
    ax.scatter([sat_mut], [mean_norm[sat_idx]], color=VARIANT_COLORS[cv],
               edgecolors='black', linewidths=1.2, marker='o')

ax.axvline(local_sat_mean, color='black', ls='--', lw=2,
           label=f'local sat (mean time_to_46 = {local_sat_mean:.0f})')
ax.axhline(95, color='gray', ls=':', lw=1, alpha=0.7)
ax.set_xlabel('Mutation index')
ax.set_ylabel('% of final CGC keys discovered')
ax.set_title('Normalized CGC saturation overlay – V5 paired seeds (n=5)')
ax.legend(fontsize=8, loc='lower right')
ax.grid(alpha=0.3)
ax.set_xlim(0, N_MUTATIONS)
ax.set_ylim(0, 105)
show_plot(fig, PLOTS / 'd1b_saturation_overlay_v5_norm.png')

```

Local saturation anchor (mean time_to_46): 3221



AUC paired tests (decayexp vs V5)

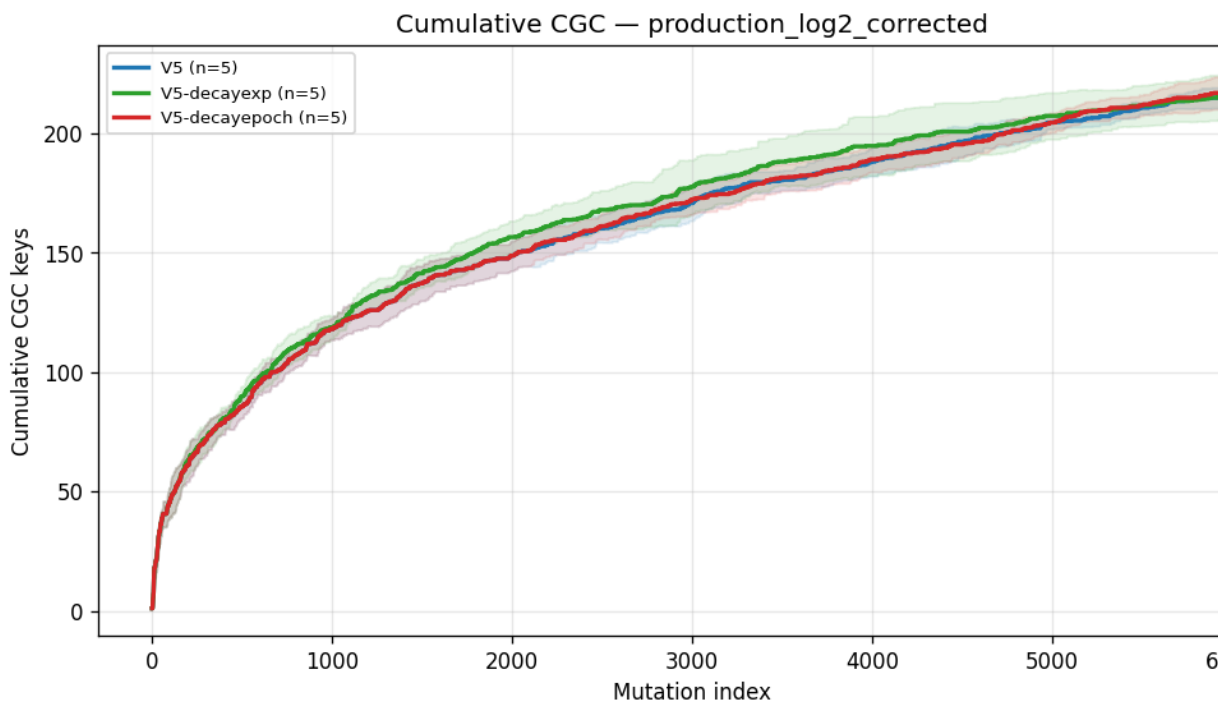
```
In [4]: auc_pt = paired[(paired.metric=='auc_normalized') & (paired.comparison=='
display(auc_pt[['cgc_variant', 'mean_a', 'mean_b', 'mean_diff', 'p_value']]).r
```

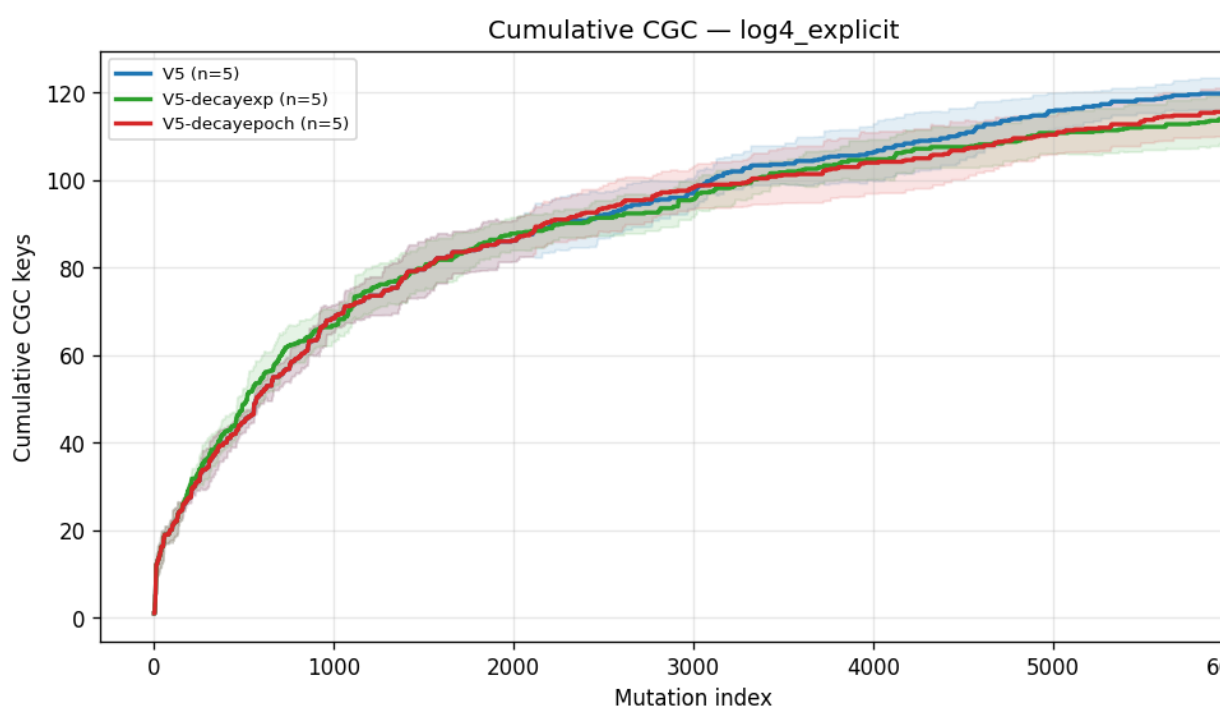
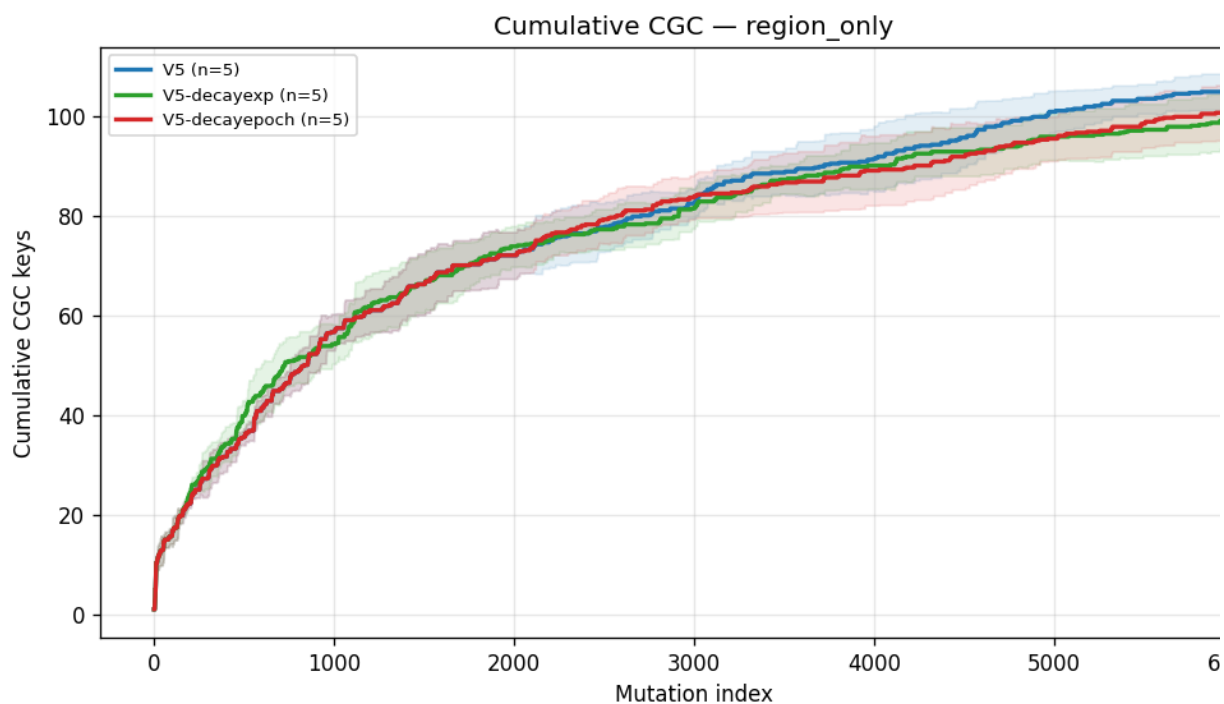
	cgc_variant	mean_a	mean_b	mean_diff	p_value
60	page_class	0.7788	0.7536	0.0253	0.0481
24	region_only	0.7683	0.7423	0.0259	0.0484
6	production_log2_corrected	0.7673	0.7445	0.0228	0.0588
42	log4_explicit	0.7854	0.7620	0.0234	0.0591

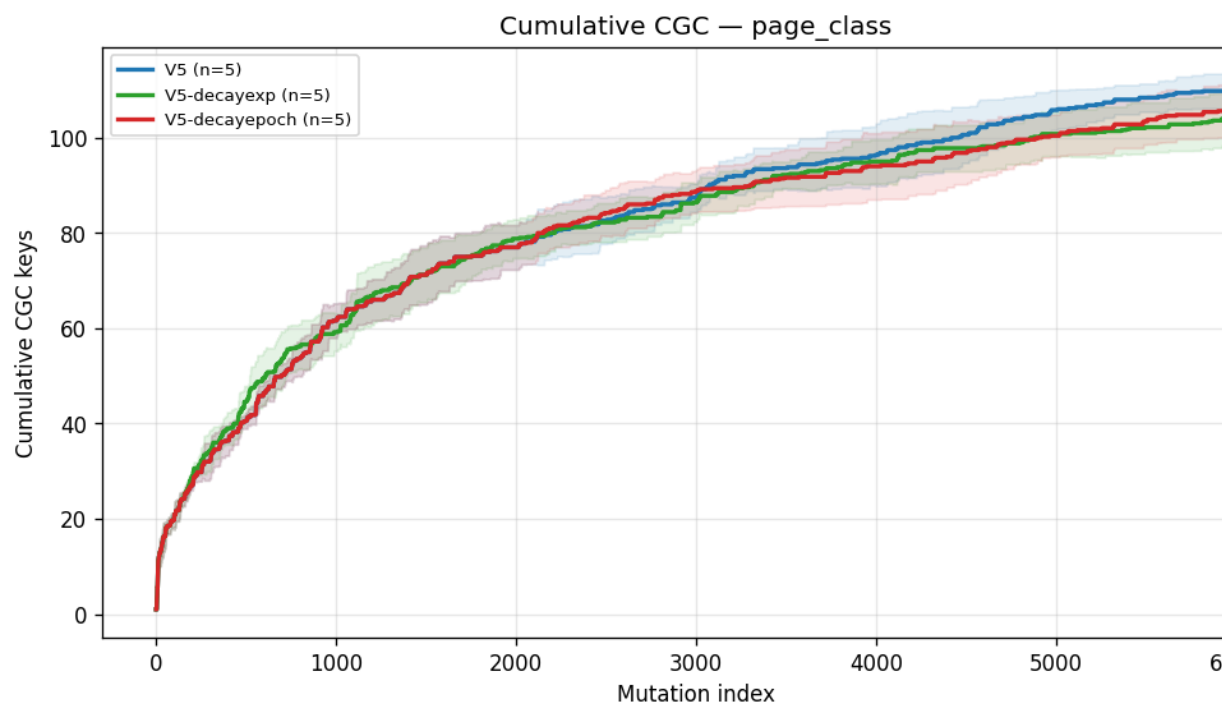
Cumulative CGC curves (1 plot per variant, paired seeds)

```
In [5]: paired_seeds = summary['paired_seeds']
x = np.arange(1, N_MUTATIONS + 1)

for cv in VARIANTS:
    fig, ax = plt.subplots(figsize=(10, 5))
    for corpus in ('V5', 'V5-decayexp', 'V5-decayepoch'):
        sub = metrics[(metrics['cgc_variant']==cv) & (metrics['corpus_var']
        curves = [cgc_curve(p, cv) for p in sub['db_path']]
        arr = np.array(curves)
        ax.plot(x, arr.mean(0), color=COLORS[corpus], lw=2, label=f'{corpus}')
        if len(curves) > 1:
            ax.fill_between(x, arr.mean(0)-arr.std(0,ddof=1), arr.mean(0)+arr.std(0,ddof=1),
                            color=COLORS[corpus], alpha=0.12)
    ax.set_title(f'Cumulative CGC — {cv}')
    ax.set_xlabel('Mutation index')
    ax.set_ylabel('Cumulative CGC keys')
    ax.legend(fontsize=8)
    ax.grid(alpha=0.3)
    show_plot(fig, PLOTS / f'cgc_curve_{cv}.png')
```







Paired tests (cgc_final)

```
In [6]: pt = paired[paired['metric']=='cgc_final'][['cgc_variant','comparison'],'n']  
display(pt.round(3))
```

	cgc_variant	comparison	n_paired	mean_a	mean_b	mean_diff
0	production_log2_corrected	V5-decayexp vs V5	5	215.6	216.2	-0.6
1	production_log2_corrected	V5- decayepoch vs V5	5	218.0	216.2	1.8
2	production_log2_corrected	V5-decayexp vs V5- decayepoch	5	215.6	218.0	-2.4
18	region_only	V5-decayexp vs V5	5	99.2	104.8	-5.6
19	region_only	V5- decayepoch vs V5	5	100.8	104.8	-4.0
20	region_only	V5-decayexp vs V5- decayepoch	5	99.2	100.8	-1.6
36	log4_explicit	V5-decayexp vs V5	5	114.2	119.8	-5.6
37	log4_explicit	V5- decayepoch vs V5	5	115.8	119.8	-4.0
38	log4_explicit	V5-decayexp vs V5- decayepoch	5	114.2	115.8	-1.6
54	page_class	V5-decayexp vs V5	5	104.2	109.8	-5.6
55	page_class	V5- decayepoch vs V5	5	105.8	109.8	-4.0
56	page_class	V5-decayexp vs V5- decayepoch	5	104.2	105.8	-1.6

Saturation profile

In [7]: `display(saturation)`

	cgc_variant	n_paired_seeds	saturation_bin_lo	saturation_bin_hi
0	production_log2_corrected	5	3300	3400
1	region_only	5	1200	1300
2	log4_explicit	5	1700	1800
3	page_class	5	1200	1300

Reproducibility


```
In [8]: display(Markdown('```\n' + manifest + '```'))  
display(Markdown(f"git commit: `{summary['git_commit']}'`"))  
display(Markdown(f"sanity_pass: **{summary['sanity_pass']}**"))
```

49aa8e0e9ae219f4e7c586fe3ffa3d3989399d8e2ee56438103da0b65e4343fb

d1b_metrics_table.csv

ce4300f7a90cb66ab8877bf460b23b9e8b4a37ea15c5df4c37695bb2f115ce3c

d1b_paired_tests.csv

6c55062f19fa36f68b00984991d3320a38b265e42fb2ee2d99c0a909e7b0b196

d1b_saturation_profile.csv

74a6a19ffe9a8dc3ceb451d1802b8f9f9beb91227819bc90c0740e705ea23eec

d1b_build_summary.json

git commit: 4fce6649946300729937595a66543afda53a1f3d

sanity_pass: **True**